

GOOGLE CYBERSECURITY CERTIFICATE

Brute Force Attacks

Google Cybersecurity Professional Certificate · Study Guide

An overview of how brute force attacks work, the tools commonly used to run them, and the defensive controls — hashing/salting, MFA, CAPTCHA, and password policy — that make them impractical against a well-defended target.

Track	Google Cybersecurity Professional Certificate
Course	Assets, Threats, and Vulnerabilities
Focus	Credential attacks & defenses
Analyst	Chayan Panchal

1 What Brute Force Attacks Are

A brute force attack is a trial-and-error method for guessing login credentials by systematically trying every possible character combination. Doing this by hand could take years; automated tooling does it in seconds to minutes depending on password complexity and the defenses in place.

Common tools

Tool	Primary use
Aircrack-ng	Wi-Fi network vulnerability testing
Hashcat	Password hash cracking
John the Ripper	Multi-format password cracking
Ophcrack	Windows password cracking via rainbow tables
THC Hydra	Online service credential brute-forcing

NOTE The same tools security professionals use to test their own systems' password strength are what attackers use offensively — the difference is authorization.

2 Prevention Measures

Hashing and salting

Hashing converts a password into a unique fixed-length value that cannot be reversed back to the original password. Salting adds random characters to a password before hashing, which increases hash complexity and defeats precomputed dictionary/rainbow-table attacks.

Multi-Factor Authentication (MFA)

MFA requires identity verification in two or more ways — a password plus an OTP, fingerprint, or hardware key. Even a successfully brute-forced password is useless to an attacker without the second factor.

CAPTCHA

Completely Automated Public Turing test to tell Computers and Humans Apart — a challenge automated scripts can't easily solve, which blocks bots from submitting login attempts at scale.

Password policy

Control
Enforce minimum length and complexity
Lock accounts after N failed attempts
Require periodic password changes
Disallow reuse of recent passwords

3 Quick Reference



Brute force — quick reference

Common brute force tools: Aircrack-ng, Hashcat, John the Ripper, Ophcrack, THC Hydra

Defenses:

hashing + salting	-> stored credentials can't be reversed or precomputed against
MFA	-> a cracked password alone isn't enough to authenticate
CAPTCHA	-> blocks automated login attempts at scale
lockout policy	-> N failed attempts locks the account